Udemy ML, DL, NLP course

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Sec-4 S Docker

Lec 1 : Dockers & Containers

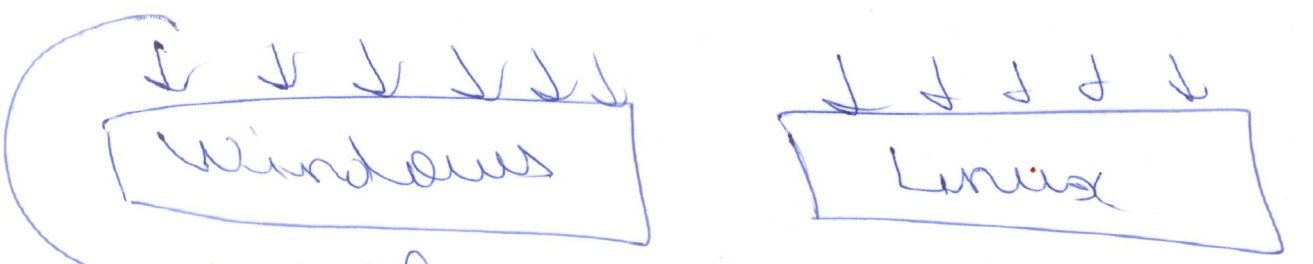
Lec 2 : Docker Image vs Containers

Lec 3 Docker vs virtual machines

Why Containers?

Suppose, we're working in Windows machine

Data science Application



→ All installation of dependencies

But may be some other
actions may not work
due to libraries mismatch

Now suppose after this,
the developer then
resolves the problem,

↳ but then we
try to start the
container.

↳ and again we
face the same problem.

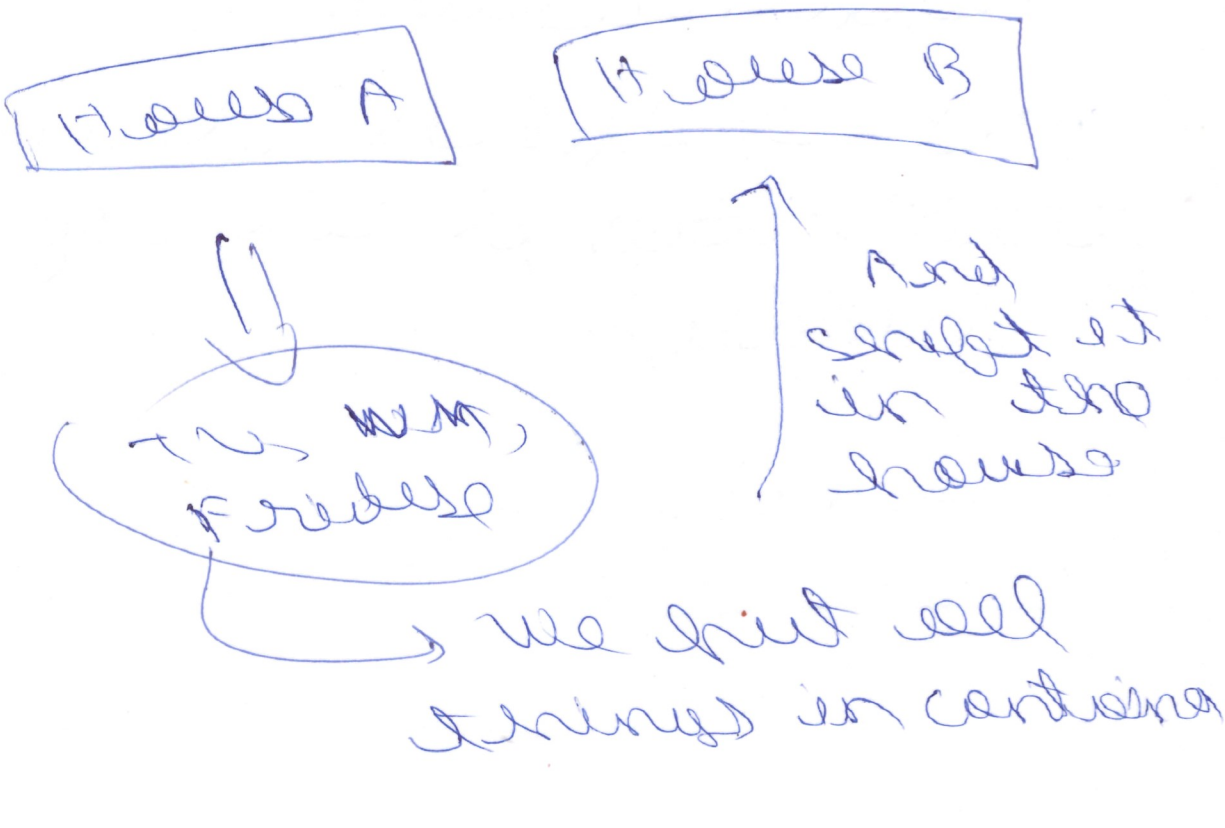
Hence to resolve this
issue & deploy applica-
tions, we use
containers.

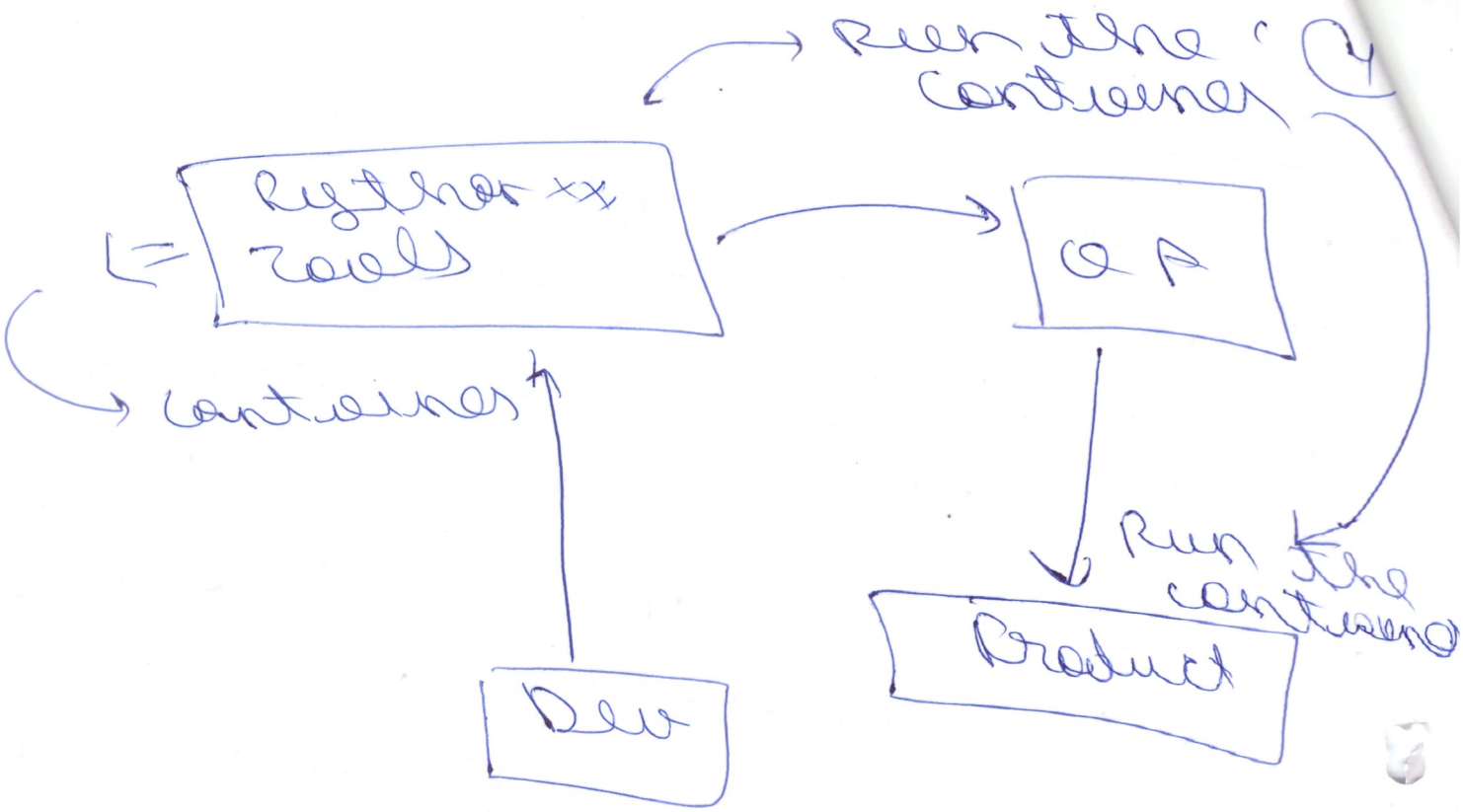
Containers

- A way to package the application with all the necessary dependencies & configuration.

- portable artifact, easily share & move this package to any environment.

- makes development & deployment more easy and efficient

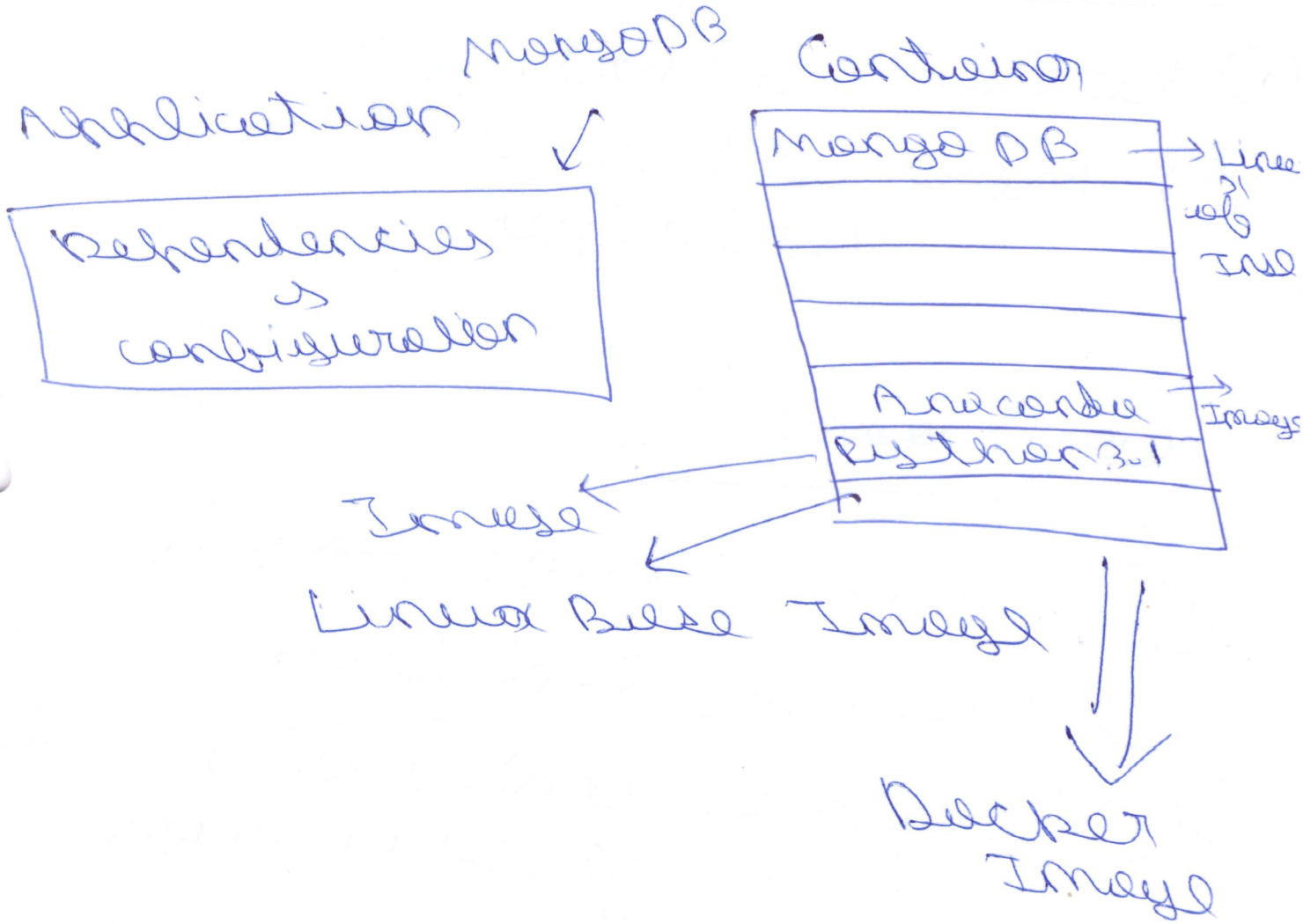




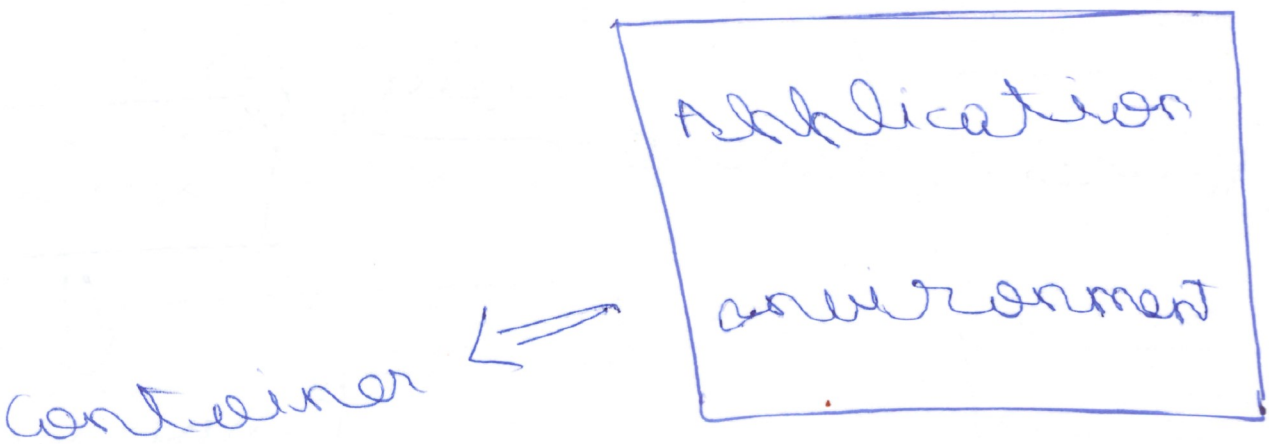
Docker :-

- It is an open platform for running & shipping application.
- It helps in delivering software & deploy it quickly.

Docker Image & Container



Docker Image → Container



Docker Image

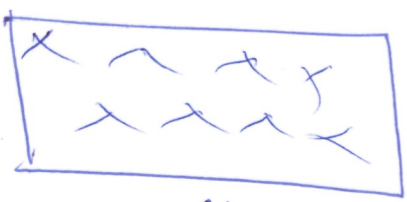
1. Docker or Artifact
2. Move or share this Artifact.

Container

1. Docker Image → Run
2. Start application
3. create a container ⇒ Environment

⇓

Running

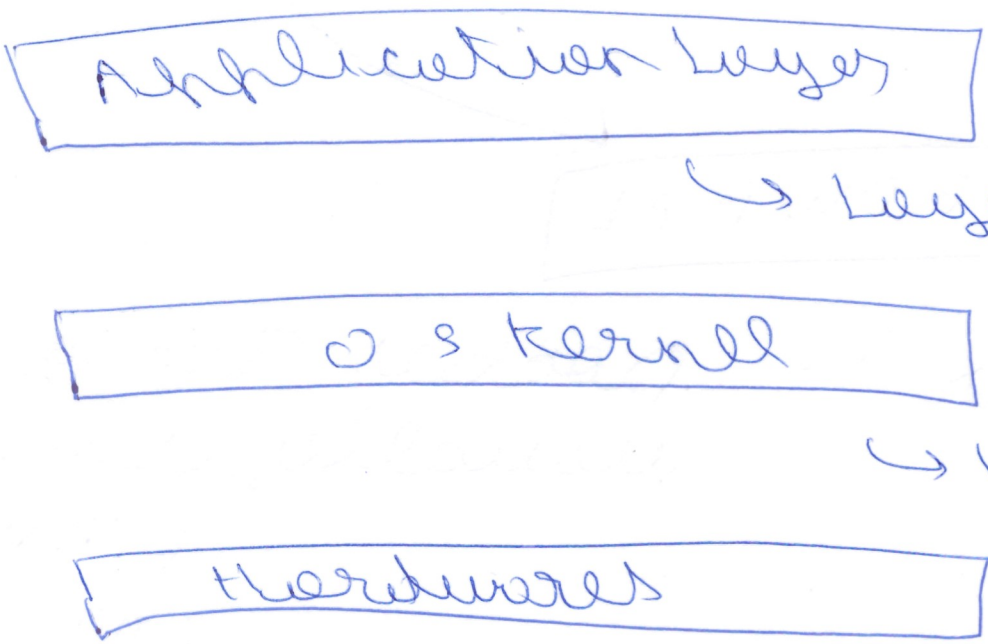
Docker Image → run → 

⇓

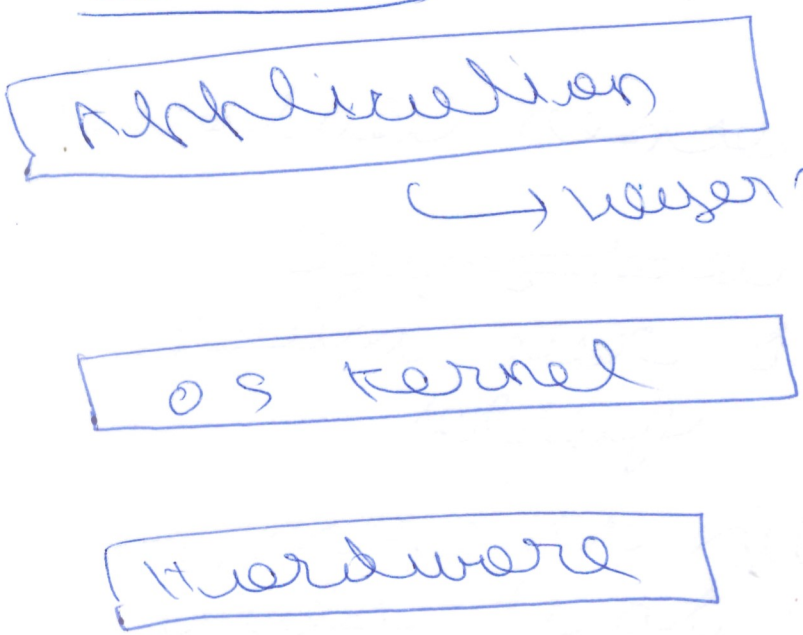
Environment

=> Docker vs Virtual machines

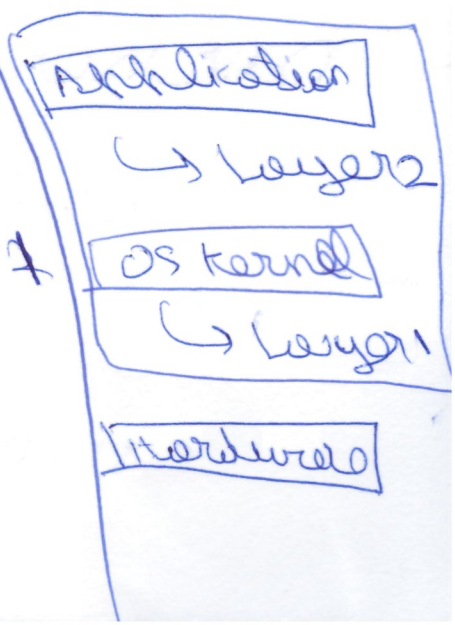
Operating System

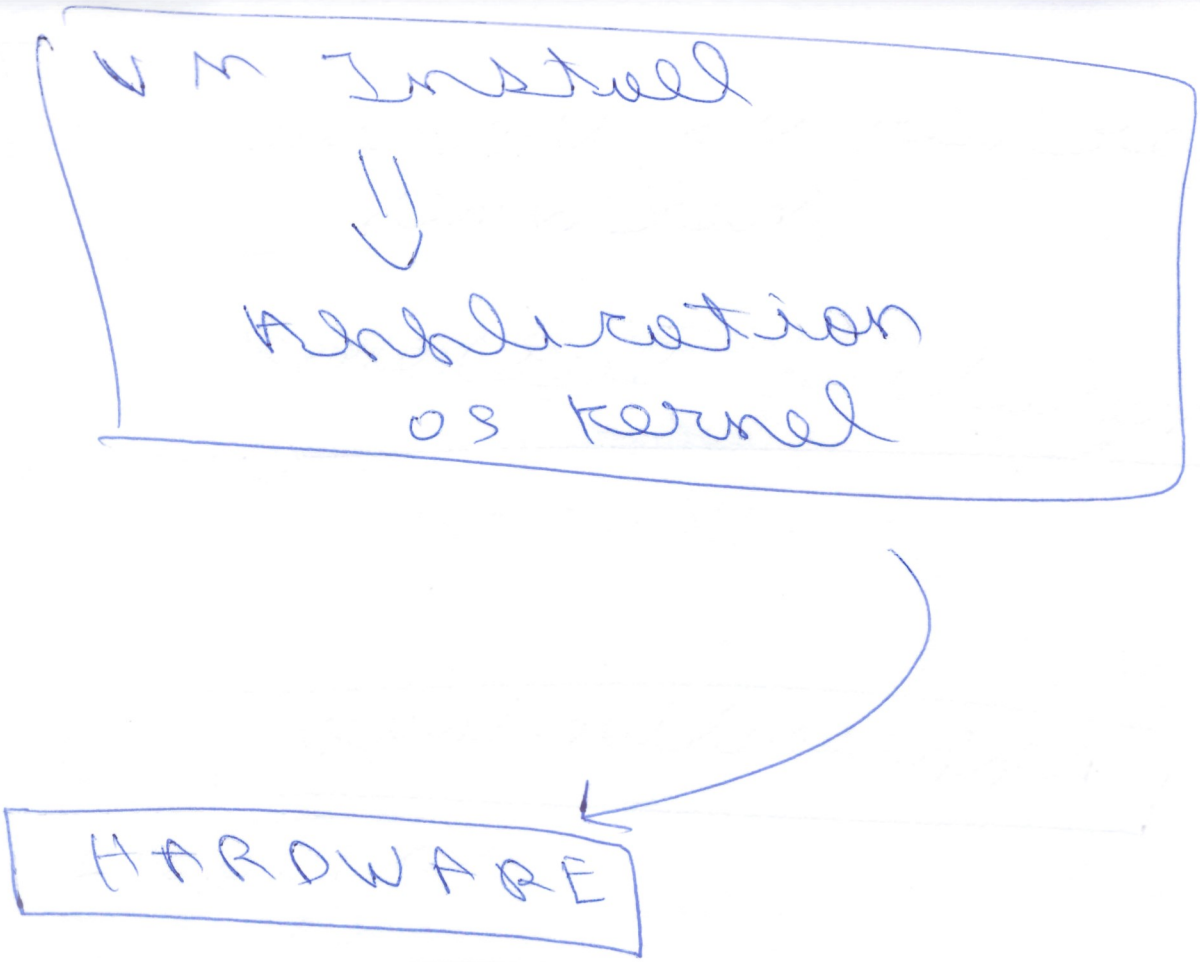


Dockers



Virtual machines





Docker :- Docker Image is usually smaller (MB)

VM :- Size will be huge (GB)

Docker :- Docker container start ~~restart~~ and run much faster

VM :- VM is correspond to it slow

Compatability :-

- Install vm on any os
- Docker Images → Compatability Issues